

EEE 419/591
Summ 2026
Additional Homework 2 – 591 Section

[Same instructions as HW 1. Changes highlighted in bold font]

Solve the following additional problem to serve as your extra work for your enrollment in the 591 version of this class. Name your file **hw2_591.py** and submit it to the designated link on Canvas. Do not add the other problems to this code. If you upload a second attempt to Canvas and it renames your file, ignore the renaming.

In this HW, only limited AI is allowed. Examples of allowed prompts on this HW:

- What is a greatest common divisor of two numbers?
- How do I write an if statement in python? Give me an example
- How do I check if a number is integer in python?
- I got this error message. How do I interpret it?

Examples of prohibited prompts on this HW:

- Write me a code that calculates the greatest common divisor of a list of numbers in python
- I ran this code. [PASTE CODE HERE]. I got this error message. [PASTE THE ERROR MESSAGE HERE]. How do I fix my code?

Do not forget to cite your AI chat if you used any. Check the Syllabus document for details.

Problem 3: Energy Calculation

Calculate the energy of the following signal

$$x(t) = e^{-t} \sin(2\pi f t) \quad , \quad t \geq 0$$

Recall that the energy of a signal $x(t)$ is given by

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

Solve for three frequency values $f = 10$ Hz and $f = 20$ Hz, $f = 10^6$ Hz.

Hint: since the signal is already decaying, you can integrate without substitution but make sure you report an accurate energy value (i.e. use large enough upper integration limit.). Print 5 decimal places.

A sample print out of your code could look like this

Energy at 10 Hz = 0.24994
Energy at 100 Hz = 0.24998
Energy at 1 MHz = 0.25000

MANDATORY – Reminder to cite AI chat